

Assignment – Java Program

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Question:

Write a Java program using ArrayList to store book titles. Add at least 5 books and search for books whose title contains a given word.

Source Code:

```
import java.util.ArrayList;
import java.util.Scanner;

public class BookSearch {

    public static void main(String[] args) {

        // Creating ArrayList to store book titles
        ArrayList<String> books = new ArrayList<>();

        // Adding book titles
        books.add("Data Structures and Algorithms");
        books.add("Operating System Concepts");
        books.add("Computer Networks");
        books.add("Java Programming");
        books.add("Database Management Systems");

        // Taking input from user
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a word to search: ");
        String word = sc.nextLine().toLowerCase();

        System.out.println("\nBooks containing the word \"" + word + "\"");

        // Searching for matching books
        for (String book : books) {

            if (book.toLowerCase().contains(word)) {
                System.out.println(book);
            }
        }
    }
}
```

```

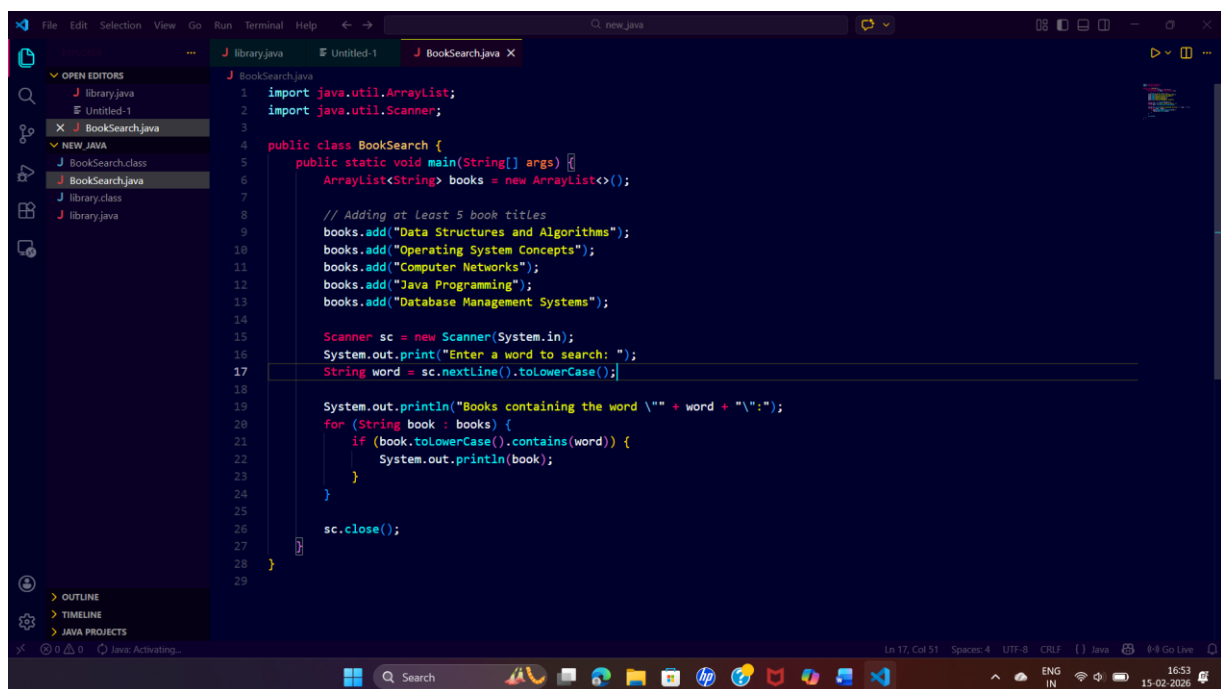
    }

}

    sc.close();
}
}

```

Source code:

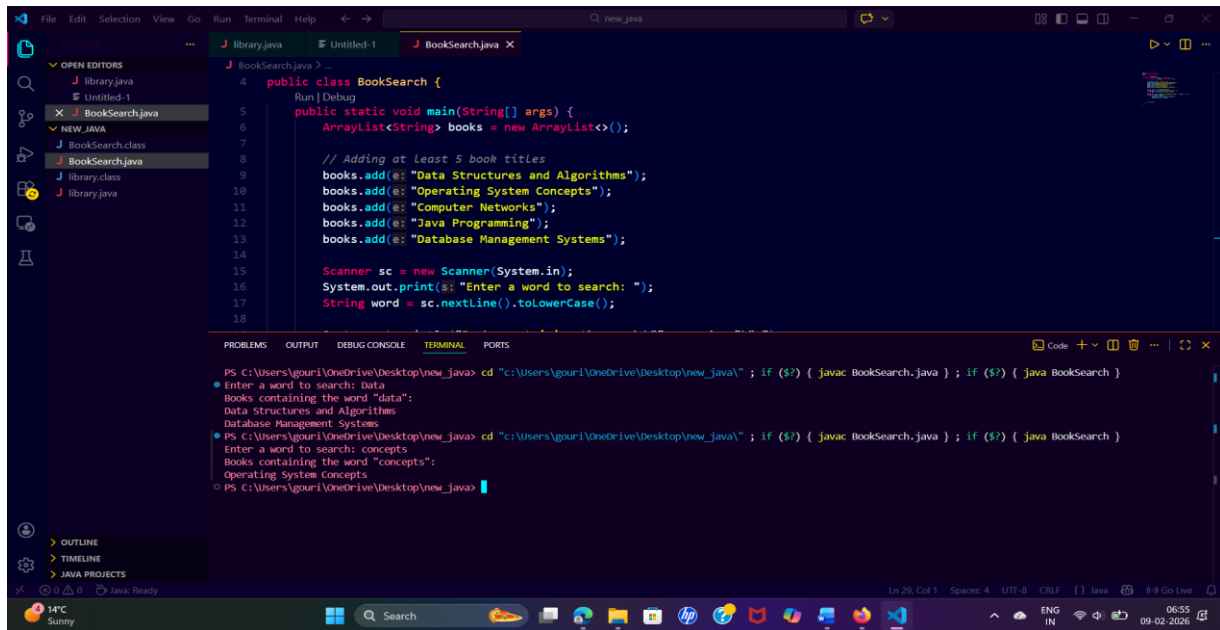


```

1  import java.util.ArrayList;
2  import java.util.Scanner;
3
4  public class BookSearch {
5      public static void main(String[] args) {
6          ArrayList<String> books = new ArrayList<>();
7
8          // Adding at Least 5 book titles
9          books.add("Data Structures and Algorithms");
10         books.add("Operating System Concepts");
11         books.add("Computer Networks");
12         books.add("Java Programming");
13         books.add("Database Management Systems");
14
15         Scanner sc = new Scanner(System.in);
16         System.out.print("Enter a word to search: ");
17         String word = sc.nextLine().toLowerCase();
18
19         System.out.println("Books containing the word \" " + word + "\" :");
20         for (String book : books) {
21             if (book.toLowerCase().contains(word)) {
22                 System.out.println(book);
23             }
24         }
25
26         sc.close();
27     }
28 }
29

```

Sample Output:



The screenshot shows an IDE with a Java project. The left sidebar displays the project structure with files like `library.java`, `BookSearch.java`, and `BookSearch.class`. The main editor shows the `BookSearch.java` file with the following code:

```
4 public class BookSearch {
5     public static void main(String[] args) {
6         ArrayList<String> books = new ArrayList<>();
7
8         // Adding at least 5 book titles
9         books.add(e: "Data Structures and Algorithms");
10        books.add(e: "Operating System Concepts");
11        books.add(e: "Computer Networks");
12        books.add(e: "Java Programming");
13        books.add(e: "Database Management Systems");
14
15        Scanner sc = new Scanner(System.in);
16        System.out.print(e: "Enter a word to search: ");
17        String word = sc.nextLine().toLowerCase();
18    }
```

The bottom panel shows the `TERMINAL` output:

```
PS C:\Users\gouri\OneDrive\Desktop\new_java> cd "C:\Users\gouri\OneDrive\Desktop\new_java\" ; if ($?) { javac BookSearch.java } ; if ($?) { java BookSearch }
Enter a word to search: Data
Books containing the word "data":
Data Structures and Algorithms
Database Management Systems
PS C:\Users\gouri\OneDrive\Desktop\new_java> cd "C:\Users\gouri\OneDrive\Desktop\new_java\" ; if ($?) { javac BookSearch.java } ; if ($?) { java BookSearch }
Enter a word to search: concepts
Books containing the word "concepts":
Operating System Concepts
PS C:\Users\gouri\OneDrive\Desktop\new_java>
```

The status bar at the bottom indicates the current line is 29, column 1, with 4 spaces, UTF-8 encoding, and CRLF line endings. The system tray shows the date and time as 06:55 on 09-02-2026.